SHAYAN AMOUZADEH

Semnan, Iran • +98 919 332 3537 • [shayanamz.7687@gmail.com] • ORCID: orcid.org/0009-0002-5518-1729
LinkedIn: linkedin.com/in/shayan-amouzadeh • GitHub: github.com/Shayan-Amz • Portfolio: shayan-amz.github.io

RESEARCH INTERESTS

- **Artificial Intelligence & Machine Learning:** Large Language Models (LLMs), Natural Language Processing (NLP), Multimodal Speech-to-Text Pipelines, and Information Retrieval.
- **Software Engineering & Computer Systems:** Automated Testing & Formal Verification, High-Performance & Deterministic Computing, Computer Graphics, and Secure Web Architectures.

EDUCATION

B.Sc. in Computer Engineering (Software Engineering Specialization) | Sep 2022 – Sep 2026
Semnan University, Faculty of Electrical and Computer Engineering | Semnan, Iran

- **Academic GPA: Selected Core CS/CE Average:** 18.03 / 20.00 (21 Courses)

  • **Final Year GPA:** 17.03 / 20.00 • **Last 2 Years GPA:** 16.93 / 20.00 • **Cumulative GPA:** 15.77 / 20.00
- **B.Sc. Thesis:** *Design & Implementation of an Intelligent Real Estate Management System*

  (**Grade: 19.50 / 20.00** • Advisor: Dr. Morteza Dorrigiv)
- **Selected Core Coursework (Grade / 20.00):** Information Technology (20.00), Programming Languages Design (19.50), Computer Networks Lab (19.50), Software Engineering (19.25), Operating Systems Lab (19.25), Web Programming (18.50), Linear Algebra (18.00), Data Structures & Algorithms (17.50).

UNDERGRADUATE THESIS & APPLIED AI RESEARCH

Developer & System Architect — Intelligent Real Estate CRM Platform

- *Advisor: Dr. Morteza Dorrigiv, Semnan University (Live: amlak-e-man.ir • github.com/Shayan-Amz/Real-Estate-File-Manager-Project) | B.Sc. Capstone Project*
- **Multi-Tenant Architecture & RBAC:** Architected and deployed a production real estate CRM (PHP/PDO, MySQL 7-table schema, 17 REST action handlers, and zero-dependency vanilla-JS PWA) featuring version-gated runtime schema migrations, composite indexing, a 15 s change-token synchronization protocol, and 3-role RBAC (*Manager / Consultant / Guest*) across 3 commercial tiers.
- **Human-in-the-Loop Voice & LLM Extraction Pipeline:** Chained browser 16 kHz mono PCM-16 WAV encoding with provider-switchable Whisper-large-v3 STT (Hugging Face, Groq, OpenAI) and an OpenRouter LLM extraction agent governed by a 16-rule Persian JSON contract, prompt-injection guardrails, a 26-field server allow-list, and client-side spoken-numeral/currency normalization into a human-reviewed modal.
- **Demand-Property Matching & Geospatial Integration:** Implemented a client-side matching engine (O(P) per demand query across active listings) using normalized bidirectional city matching, ±15% area tolerance, and 3-mode budget constraints (≤1.10×), integrated with the lazy-loaded Neshan Leaflet SDK for interactive geocoding and deep-link routing.
- **Security Hardening & Automated Evaluation:** Designed 30-hour HMAC-SHA256 tenant-bound tokens verified in constant time, per-request consultant revocation, SHA-256 masked tenant IDs in guest responses, a 150 req/min sliding-window rate limiter, dual-stage image watermarking/sanitization, and 47 automated tests alongside an offline Persian WER/F1 evaluation console secured by HS256 JSON Web Tokens (JWT, RFC 7519).

SELECTED RESEARCH & OPEN-SOURCE PROJECTS

IEEE 754 Single-Precision Floating-Point Unit github.com/Shayan-Amz/Floating-Point-ALU

C++17 Header-Only (constexpr), CMake, 4-Way Cross-Platform CI | Hardware Arithmetic Verification

- Implemented a bit-exact, compile-time (`constexpr`) C++17 software model of an IEEE 754 binary32 arithmetic datapath (add, sub, mul, div: unpack → sticky-bit shift alignment / 48-bit product / 27-bit integer long division → normalization → Guard-Round-Sticky rounding → pack) supporting all 4 IEEE 754 rounding modes, subnormals, signed zeros, ±∞, quiet/signaling NaN propagation, and bitwise trace inspection.
- Verified correctness via differential testing against the host FPU under fesetround(): executed 32,026,992 operand-pair checks (41×41 edge grid + 8 randomized stress distributions) with 0 mismatches, enforced on a 4-way GCC/Clang/AppleClang/MSVC CI matrix compiled with -frounding-math and /fp:strict.
- Quantified defects in an initial course adder using a 1.4M-sample accuracy harness (58.2% bit-exact, 40.8% 1-ULP error, 1.1% >1-ULP error) and achieved 100% bit-exact compliance by adding sticky-bit alignment, round-to-nearest-even, and gradual-underflow handling.

Deterministic Graphics Pipeline & CPU Software Rasterizer github.com/Shayan-Amz/OpenGL-Dream-Village

- *C++17, C99, OpenGL 1.1, FreeGLUT, CMake, 3-OS GitHub Actions CI | Computer Graphics & Determinism*
- Refactored an immediate-mode OpenGL 1.1 animated scene into a decoupled `AnimationState` and pure `render()` architecture compiled against both system OpenGL and custom headers for cross-platform FreeGLUT execution and headless GPU-less regression testing.
- Authored `softgl`, a C99 deterministic software rasterizer implementing 49 GL/GLUT entry points — Juan Pineda's (SIGGRAPH '88) parallel half-space edge functions in double precision, pixel-center sampling, top-left fill rule, polygon fan-triangulation, DDA line rasterization, and Gouraud shading — verified by 28 unit self-test assertions.
- Enforced pixel-exact CI regression (--max-diff-ppm 0) across 4 dynamic scenes (up to 7,017 triangles and 1.26M fragments/frame at 1300×650 resolution) on GCC, AppleClang, and MSVC, resolving an ARM64 FMA-contraction discrepancy via -ffp-contract=off and eliminating 605 per-frame legacy GL errors.

Bare-Metal ARMv7-A VGA Framebuffer Renderer github.com/Shayan-Amz/ARM-Assembly-Flag-Renderer

- *ARMv7-A GNU/Zig Assembly, Python Unicorn CPU Emulator, CI | Low-Level Systems & Emulation*
- Wrote a bare-metal ARMv7-A (Cortex-A9 / Intel DE1-SoC) graphical renderer filling the memory-mapped 320×240 RGB565 VGA framebuffer at 0xC8000000 via an AAPCS-compliant leaf routine using caller-saved registers (r0–r3) with zero stack spills, packing two 16-bit pixels per 32-bit word store across the 1,024-byte row stride.
- Built a Python Unicorn-engine emulation harness executed in CI across GNU binutils and Zig/Clang cross-assemblers to instrument every instruction and memory write, verifying exact stripe geometry (80/80/80 rows), clean halt detection, and zero out-of-bounds stores.
- Reduced executed instructions from 353,286 to 116,201 (−67% / 3.04× reduction) and eliminated 9,216 out-of-bounds memory writes and 46,080 redundant padding writes compared to the initial baseline implementation.

End-to-End Box-Office Scraping & Content Recommender Pipeline github.com/Shayan-Amz/Movie-ML-Pipeline

Python 3.10+, Pandas, BeautifulSoup4, Streamlit, Pytest & Ruff CI | Data Engineering & Content Filtering

- Engineered a `pip`-installable `src`-layout data pipeline scraping 12 years (2013–2024 / 1392–1403 SH) of Iranian box-office archives into a clean **686-film × 52-column dataset** with outlier filtering, Persian numeral normalization, and **43 multi-label one-hot genre features** (102 unique genre signatures).
- Designed a deterministic content-based recommendation engine matching exact 43-bit genre signatures to deliver dual Top-K rankings by critic rating and box-office revenue inside a cached Streamlit UI, backed by **14 pytest unit tests** and `ruff` CI checks across Python 3.10 and 3.12.

Exact-Optimal Fractional Knapsack & DP Benchmark Suite github.com/Shayan-Amz/Fractional-Knapsack-Greedy

C++17 Header-Only, Python Benchmarking, 4-Way Cross-Platform CI | Algorithmic Optimization

- Implemented an O(n log n) header-only C++17 greedy fractional knapsack solver using a division-free cross-product value-density comparator (`v_a*w_b > v_b*w_a`) with `std::stable_sort` for deterministic tie-breaking, paired with an O(nW) 1D rolling-array 0/1 dynamic programming reference solver, LP-relaxation upper bound for branch-and-bound, and a formal exchange-argument optimality proof.
- Validated optimality across **328,582 automated checks (0 failures)** — comparing against CLRS reference instances, 0/1 DP bounds, and exhaustive subset-and-fractional-item enumeration (O(n 2^n) brute-force oracle) across 20,000 randomized instances — on a 4-way GCC/Clang/AppleClang/MSVC CI matrix; benchmarked the O(n log n) greedy solver at **1.86 ms vs. 26.38 s** for the O(nW) 0/1 DP reference at n=20,000 and characterized integrality gap decay from 30.0% (n=5) to 0.05% (n=200).

PROFESSIONAL & INDUSTRY EXPERIENCE

Computer Engineer & IT Specialist Mar 2025 – Present (Full-Time)

Khakab Sazeh Ghomes Co. | Semnan, Iran

- **Enterprise IT & Network Administration:** Manage internal network infrastructure, hardware/software provisioning, and daily system administration across company operations.
- **Database & Project Automation:** Design and maintain relational databases and automated financial/engineering billing models using Microsoft Access, Excel, and Microsoft Project (MSP) to streamline project scheduling and resource tracking.

Game Developer & Software Engineering Intern Jul 2025 – Dec 2025

Sargarmi Sazan Aseman Omid (Game Development Studio) | Iran

- **Data-Oriented Architecture (ECS):** Architected core gameplay mechanics in Unity (C#) utilizing the Entitas Entity Component System (ECS) framework on proprietary studio projects, strictly decoupling immutable component state from reactive execution systems.
- **UI & Performance Engineering:** Improved runtime frame-time consistency, modularity, and maintainability over conventional OOP inheritance hierarchies; additionally built open-source Unity 6 uGUI/URP 2D interfaces with LINQ-driven reactive search and sorting (github.com/Shayan-Amz/Unity-School-Directory-UI).

TECHNICAL SKILLS

- **Programming Languages:** C++ (C++17, STL, `constexpr`), Python 3, ARMv7-A Assembly, C#, PHP 8, JavaScript (ES2017+), SQL, C99, HTML5/CSS3.

- **Systems & Verification:** IEEE 754 Arithmetic Modelling, Differential FPU Testing (`fesetround`), Deterministic Software Rasterization, Unicorn CPU Emulation, Bare-Metal Memory-Mapped I/O (DE1-SoC VGA).
- **AI, NLP & Data Pipelines:** LLM Structured Extraction (OpenRouter, Prompt-Injection Guardrails), Whisper STT Audio Processing, Persian NLP Normalization, Pandas, BeautifulSoup4, Streamlit, WER/F1 Evaluation.
- **Web, Databases & Security:** RESTful APIs, Zero-Dependency PWA/SPA, Neshan Leaflet SDK, MySQL / MariaDB / SQLite (PDO, Schema Migrations), Stateless HMAC-SHA256 & HS256 JWT Auth (RFC 7519), Rate Limiting.
- **Graphics, Game Engines & DevOps:** OpenGL / FreeGLUT, Unity 6 (uGUI, URP 2D, LINQ), Entitas ECS, GitHub Actions (Multi-OS Matrix CI), CMake, GCC / Clang / MSVC, Docker, Pytest, Ruff, Puppeteer.

LANGUAGES

- **Persian:** Native Proficiency

  • **English:** Advanced Working Proficiency (IELTS Academic in preparation).